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Letters from an Astrophysicist

by Neil deGrasse Tyson

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123 Highlights | 7 Notes

Highlight (Yellow) | Location 138

Hope

Highlight (Yellow) and Note | Location 152

yes, the universe wants to kill us. But on the other hand, we all want to live. So let's find a way together to deflect the asteroids, find the cure to the next lethal virus, mitigate hurricanes, tsunamis, volcanoes, etc. This can only be enabled by the efforts of a scientifically and technologically literate public. Therein lies a hope on Earth far greater than ever promised by the act of prayer or introspection.

On hope and optimism...

Highlight (Yellow) and Note | Location 172

people who fail in life are those whose ambitions were insufficient to overcome all the forces that work against them. And yes, failure is common to us all. But ambitious people use their failures as lessons to heed, as they push forward toward their goals. Don't fear change. Don't fear failure. The only thing to fear is loss of ambition. But if you've got plenty of that, then you have nothing to fear at all. Good luck on your journey, as I offer you the opening quote in my memoir, *The Sky Is Not the Limit** Beyond the judgments of others Rising high above the sky Lies the power of ambition All the best, on Earth and in the universe. Neil

On fear...

Highlight (Yellow) and Note | Location 227

IQ correlates nicely with GPA in high school and college, but after your first job, nobody ever asks what your college GPA was. What matters are your people skills, leadership skills, real-world problem solving skills, integrity, business acumen, reliability, ambition, work ethic, kindness, compassion, etc. So for me, conversations about race and IQ are of no practical consequence, any more than are conversations about race and hair color, or race and food preferences.

On IQ...

Highlight (Yellow) | Location 260

when money is the sole carrot, people can lose sight of life's deeper sources of happiness. Until you major in physics or astrophysics in college (taking all the attendant math courses) you will not know for sure what you're better at—academics or sports.

Highlight (Yellow) | Location 276

the expectation that everyone else in our culturally pluralistic land should hold exactly your own outlook, on all issues.

Highlight (Yellow) and Note | Location 278

When empowered by this state of mind, objective realities matter. These are the truths of the world that exist outside of whatever your belief system tells you.

Scientifically literate

Highlight (Yellow) | Location 293

The Carl Sagan dictum “extraordinary claims require extraordinary evidence” remains a potent guide when investigating the natural world for its underlying order. But it comes with a recurring risk: knowing enough about a subject to think you are right, but not enough about the subject to know you are wrong.

Highlight (Yellow) | Location 302

the weight of the evidence is incommensurate with the strength of the claims.

Highlight (Yellow) | Location 329

“cover-up” or “conspiracy” is the battle cry of people who want to believe, in the face of insufficient data to fully support their claims.

Highlight (Yellow) | Location 333

Once you confess to not knowing what you are looking at, no logical line of reasoning allows you to then declare that you know what you are looking at. And that includes assertions that the flying shapes “must be” intelligent, technologically advanced aliens from distant planets secretly observing the behavior of Earthlings. You simply bear insufficient evidence to make that jump, however tempting it may be. A similar argument from ignorance comes from the Big Bang. When I am asked what was around before the Big Bang, I say, “We do not yet know.” Often the reply is, “It must be something—it was surely God.” To go from “We don't know” to “It must be God” is another example of an argument from ignorance. This kind of disconnect has no place in rational investigations, yet it perennially permeates the thoughts and statements of people who already know what they want to believe.

Highlight (Yellow) | Location 345

The day real evidence comes forth, you will not need congressional hearings parading high-ranking eyewitnesses on the subject. Until any of that happens, UFO sightings of category (4) are simply intriguing unidentified lights and shapes in the sky—perhaps worthy of further study like any mystery in science—but without conspiracy theorists invoking cover-ups to bridge all gaps in the data, convincing themselves of what they are already sure is true.

Highlight (Yellow) | Location 395

the end of what Mayan scholars call the “long count,” begun on August 11, 3113 BC, the beginning of the universe according to Mayan calculations. At the end of the long count, they imagined the end of the universe. With regard to the beginning of the universe, they were off by at least 13 billion years. So there’s no reason to think they will get the end correct. None of them mention the year 2012. And all the doomsday comet predictions via Nostradamus apply to the year 2000. Of course, none of those apocalyptic predictions came true.

Highlight (Yellow) | Location 404

the current Gregorian calendar used worldwide is accurate to a day in about 44,000 years. No other calendar comes close to this.

Highlight (Yellow) | Location 417

As for ancient civilizations, if you wish to go back 5,000 years to cite behavior you wish to emulate, then consider some other baggage that comes with it—the worship of cats, the divinity of the Pharaohs, the obsession with expensive, overbuilt, triangular tombstones. And while we are at it, why not include the Aztecs? Time to rip the pulsing heart out of virgins to appease the gods. Or time to eat the flesh of those you conquer, to make you stronger. And to complete the scene, why not die of disease and pestilence before you turn 40? Like so many other attributes, astrology was not an achievement of these civilizations, it was a cultural liability. Keeping it real, Neil deGrasse Tyson

Highlight (Yellow) | Location 436

“What can be more palpably absurd than the prospect held out of locomotives traveling twice as fast as stage-coaches?” —The Quarterly Review, 1825 “Men might as well project a voyage to the moon as attempt to employ steam navigation against the stormy north Atlantic Ocean.” —Astronomer Dionysius Lardner, 1838 “Man will not fly for 50 years.” —Wilbur Wright to his brother Orville, 1901

Highlight (Yellow) | Location 444

these quotes, in the end, are about perceived limits to our technology and not the laws of physics themselves,

Highlight (Yellow) | Location 461

the laws of gravity are there and work every time, no matter who is doing the experiment, no matter what apparatus you use to measure it, and they work whether or not you choose to believe in them.

Highlight (Yellow) | Location 464

ghosts, apparitions, spirits, etc. None of these claims survive laboratory scrutiny. They all simply go away under controlled circumstances.

Highlight (Yellow) | Location 505

In the era of “modern” experimental science, which is basically traceable from Galileo and Sir Francis Bacon# onward (the past 400 years), there is tested science that has achieved consensus and there is frontier science. The frontier science changes monthly, if not weekly, awaiting good enough data to resolve the controversies. The tested science, that for which consensus of observations and experiment has emerged, does not change. What can, and often does, happen is that new ideas extend the range of previously tested ideas, but it does not undo them.

Highlight (Yellow) | Location 571

the claims are useless to the research scientist. FYI: “useless” is not the same word as “false.” Until somebody produces biological tissue (even big-foot poop would be a start) from which DNA can be extracted, there is not much a biologist can do with the claims. If you feel confident that there are eight-foot, undocumented, prehistoric apes running around the Pacific Northwest, you should mount expeditions to find them—you don’t have to kill it; just capture one. Your efforts will be better spent in search of useful evidence than in trying to convince people of what you think is true in the absence of it.

Highlight (Yellow) | Location 651

wait, there’s more . . . How about the fact that the entire universe is composed of just 92 elements? Or that there are only four fundamental forces of nature? (strong, weak, electromagnetic, gravity) Or that there are only four classes of fundamental particles? (quarks, electrons, neutrinos, and photons) Or that nearly all behavior of electromagnetic waves (light) can be derived from a set of four equations that all fit on a Post-it® note? So you can be awed by the complexities manifest the world, or you can instead be astonished at how simple it all is.

Highlight (Yellow) | Location 661

One of the great challenges of cosmic discovery, and discovery in general, is knowing the difference between when things look the same and when they are the same.

Highlight (Yellow) | Location 667

When Sir William Herschel, in the 1800s, first saw dots of light that moved slowly across the sky, he knew they could not be stars, but they looked like stars through his telescope, so he called them “star-like,” which in Latin becomes “aster-oid” or just asteroids.

Highlight (Yellow) | Location 676

best instead to ask, “What is the thing?” rather than “What does the thing look like?”

Highlight (Yellow) | Location 685

I have an unorthodox philosophy regarding root-finding. I just don’t care. And that’s not a passive absence of caring it’s an active absence of caring. Since any two people in the world have a common ancestor—depending on how far back you look, the line we draw to establish family lineage is entirely arbitrary. When I wonder what I am capable of as a human being, I don’t look to “relatives,” I look to all human beings. That is the genetic relationship that matters to me. The genius of Isaac Newton, the courage of Joan of Arc and Gandhi, the athletic feats of Michael Jordan, the oratorical skills of Sir Winston Churchill, the compassion of Mother Teresa. I look to the entire human race for inspiration for what I can be—because I am human. I don’t care if I am a descendent of kings or paupers, saints or sinners, the brave or cowardly. My life is what I make of it. So I respectfully decline your invitation, but I do so knowing that ever since Alex Haley, most people find this pastime to be immensely enlightening.

Highlight (Yellow) | Location 702

nobody says Earth was formed in 4.6 billion years BC. We simply say it formed 4.6 billion years ago. Same holds true for geological and biological time-keeping.

Highlight (Yellow) | Location 704

2.The origin of Earth spanned at least a hundred million years. So a precise date and time to start a cosmic calendar would be meaningless—like attempting to celebrate the nanosecond (billionth of a second) that you were born. The time it took you to exit the birth canal vastly exceeded this measure of things.

Highlight (Yellow) | Location 711

The Gregorian calendar is, quite simply, the most accurate and stable calendar ever devised. The Jesuit priests, appointed by Pope Gregory back in the 16th century, did a bang-up job in their calculations.

Highlight (Yellow) | Location 722

Furthermore, as an atheist, you surely use the words: “holiday” and “goodbye,” yet their origins come from “holy day,” and “god be with you.”

Highlight (Yellow) | Location 734

I am about an hour north of Baghdad. This is a place in your book that has gotten a few references.‡ I am able to talk to some of the local people who are well aware of their past significance in the role of the sciences and give them opportunity to speak to me of many other things that I was previously unaware of, that have transpired right here in my temporary back yard. These conversations, that are spawned from the knowledge imparted by

your book, leave me feeling more like a heavily armed tourist, rather than an occupying invader. A book like yours, I imagine, leaves people wanting to gaze upon the stars and reflect on some of what they have just read. I have discovered that the night sky is teeming with a lot more than I ever thought possible, when I turn on my Night Vision Goggles. How many of your readers can say that they were inspired to use defense technology to relax after a long day at work?

Highlight (Yellow) | Location 742

inspired me to think, to use my head for something other than just an ear spacer!

Highlight (Yellow) | Location 752

Baghdad has deep history in the sciences, especially in mathematics—algebra in particular. Furthermore, next time you look up at night (as you may recall from the book), two-thirds of all stars that have names, are named in Arabic, enabled by major advances in navigation a thousand years ago. The most enduring thing about being human is the discovery of cosmic truths that transcend culture, politics, religion, and time, forming the corpus of knowledge and wisdom that we call civilization.

Highlight (Yellow) | Location 773

Lucy in the Sky with Diamonds Wednesday, June 10, 2009 I'm Georgette Burrell and I'm seven years old. I saw your special on how Pluto is a dwarf planet. I thought that it was very cool. I heard of a planet (or star) called Lucy that is a big diamond. My question is how would scientists know about what it is if it's so far away? Thanks, Georgette Excellent question, Georgette. Many dead stars are made of carbon (they are white dwarfs). Under high pressure, pure carbon turns into diamond. These stars have strong gravity, which puts their carbon under high pressure. So we can use math to calculate the star might just be made of pure diamond. Neil

Highlight (Yellow) | Location 796

In Earth orbit, you already possess half the energy to get anywhere else in the solar system. In other words, the energy necessary to achieve Earth orbit is exactly half the energy it takes to leave Earth entirely. Slingshot maneuvers are for spacecraft that do not launch with enough fuel to reach their destination. They take longer than ballistic trajectories because their total distance traveled can be twice what it would otherwise be, as they fall towards the planets and moons that would provide the needed gravity boost.

Highlight (Yellow) | Location 800

How fast could we have them going? Is 39,000 mph the realistic limit these days? Any speed at all. It's just a matter of fuel available to accelerate and decelerate. Escape velocity for Earth is 25,000 mph. That should get you to Iapetus in 10 years.

Highlight (Yellow) | Location 831

Dear Chris: Disney's 1979 film The Black Hole. Worst ever—given how fertile the scientific material was—until the 1998 film Armageddon, which violates more laws of physics (per minute) than any other film in the universe. Neil

Highlight (Yellow) | Location 838

the board reads “Rhinoviruses are among the major causes of the common cold. They consist of DNA surrounded by a protein coat.” However, a rhinovirus contains RNA (ribonucleic acid) and not DNA (deoxyribonucleic acid) as the board states. We acknowledge that this may seem trivial and do not mean to sound nitpicky, but whether or not a virus contains DNA or RNA is actually one of the most fundamental ways in which viruses are classified and distinguished. It affects the viruses’ mode of transmission, mode of replication, stability, and physical characteristics among other core properties. Therefore, we decided this was important enough to write to you about.

Highlight (Yellow) | Location 858

Unlike the near-circular orbits of planets, most comets travel on highly elongated trajectories, crossing other people’s orbits as they plunge in and out of the inner solar system. Made principally of ice, as a comet nears the Sun’s heat its outer layers evaporate, creating an enormous ball of reflective gas—the “coma.” These gases also stretch through interplanetary space, forming the celebrated comet’s “tail.”

Highlight (Yellow) | Location 874

Fibonacci (ca. 1170 – ca. 1250), Italian (Pisa) mathematician, most famous for the number series that bears his name, where each entry is the sum of the two previous entries; i.e., 1, 1, 2, 3, 5, 8, 13, 21, 34, etc.

Highlight (Yellow) | Location 879

During the Golden Age of Islam of a millennium ago, the intellectual center of the world was Baghdad.

Highlight (Yellow) | Location 1025

6/10 of a penny of your tax dollar. That is the entire allocation to NASA. That is the cost of your access to the universe

Highlight (Yellow) | Location 1048

For example, nowhere in the Bible is Earth described as a three-dimensional object. All references to it are flat. And until the 15th century, so too were all maps of the world, informed by scriptures. We can celebrate the cultural history of this notion, but it’s objectively false.

Highlight (Yellow) | Location 1050

Same holds for the value of Pi. In the Bible, a passage (I Kings VII) can only be true if Pi exactly equals 3.0. But we know better than this (so too did the earlier Babylonians, who calculated that Pi was a number between 3 and 4). But just because the Bible says $\text{Pi} = 3$, does not mean $\text{Pi} = 3$.

Highlight (Yellow) | Location 1052

The fact that the people who wrote the Bible made $\pi = 3$ and Earth a flat disc is of some historical interest and worthy of study in history class, philosophy class, or religion class. But it has no place in the field of science, whose goal is to find truths of the universe that sit independently of opinion.

Highlight (Yellow) | Location 1056

there is no tradition of scientists knocking down the doors of Sunday School telling the preachers what to teach. Scientists don't picket outside of churches or shoot people who enter them. Scientists have no tradition of heckling preachers during sermons.

Highlight (Yellow) | Location 1072

religion class should cover all the world's faith-based philosophies and belief systems. I think, historically, such a class was omitted because religions themselves are not fans of tolerating other religions.

Highlight (Yellow) | Location 1089

a-gnostic: A term coined by the nineteenth-century naturalist Thomas Henry Huxley in reference to a person who claims neither faith in, nor disbelief in, God. Today, it references a person who allows the possibility of a God, but remains skeptical.

Highlight (Yellow) | Location 1112

trends in the data remain the primary commodity of what is true in science, and not the impassioned testimonies of scientists themselves.

Highlight (Yellow) | Location 1136

Four billion years ago, cyanobacteria transformed Earth's atmosphere by making O_2 in the greatest ecological disruption in the history of life on Earth, killing all surface-living anaerobic bacteria. Human-induced global climate change is (at the moment) not unstoppable. And, of course, the solution will come from science and technology—via enlightened leadership. Just as the problem arose from science and technology—via shortsighted leadership. But this cycle is nothing new. We solved the food shortage problem in the world*—which was as big a fear at the end of the 19th century as global warming has been in the 21st century.

Highlight (Yellow) | Location 1164

Science follows evidence. And when strong evidence supports an idea, the concept of belief, when invoked the way religious people use the word, is unnecessary. In other words, established science is not an ensemble of beliefs, it's a system of ideas supported by verifiable evidence.

Highlight (Yellow) | Location 1168

Evolution by natural selection is a non-controversial tenet of modern biology. It's not controversial among biologists, that is. Among religious fundamentalists, biological evolution does not square with their faith-based belief system, in which they invoke the Bible as an inerrant understanding of the physical world. This leads to claims, for example, that Earth is no more than 10,000 years old. And that there was a literal flood in which all of Earth was covered in water. No evidence supports this. Not only that, all evidence argues against it. And so one is left "believing in" stories that are demonstrably false.

Highlight (Yellow) | Location 1223

Natural selection drives evolution. And evolution is occurring all around us. All the time. It's best seen among species that have fast reproductive cycles, where variations can be selected and identified on time scales short compared with a human life. The bacterial branch of the tree of life is huge—much larger in variation than what is found among vertebrates, for example. Among bacteria as well as viruses, we see speciation all the time.

Highlight (Yellow) | Location 1228

Not all species are evolving at all times. For example, the Coelacanth (pronounced See-luh-kanth) is an extremely successful bottom-feeding fish that has not changed substantially for the past 360 million years. Horseshoe crabs go back even further—450 million years. If you are a successful species, there is no driver to influence change.

Highlight (Yellow) | Location 1231

mammals have changed dramatically over the past 65 million years. When I say dramatically, I mean visually, not biologically. We share more than 90% identical DNA with all mammals, even mice.

Highlight (Yellow) | Location 1232

Among mammals in the tree of life is the branch called primates, such as lemurs, monkeys, and the great apes, including humans. It's commonly thought that humans evolved from monkeys. But this is not true. We all have a common ancestor. The ape that is closest to us is the chimpanzee. In other words, chimps and humans have a relatively recent common ancestor.

Highlight (Yellow) | Location 1236

Contrary to your assertions that chimps and humans are completely different from each other, chimps and humans have every muscle and every bone in common. Chimps and humans even have the same facial expressions. But most importantly, we have only trifling differences in our DNA.

Highlight (Yellow) | Location 1260

To keep our identity secret will be very hard, now that our radio bubble is more than 70 light-years out there, and expanding.

Highlight (Yellow) | Location 1274

science seeks all three (truth, understanding, meaning), but primarily concerns itself with obtaining sufficient knowledge of how the universe works to be able to make testable predictions about its past and future behavior. Computer simulations can substitute for past and future behavior when sensible.

Highlight (Yellow) | Location 1301

Why does the moon always show the same face to Earth?

Highlight (Yellow) | Location 1304

Why is August hotter than June, when the Sun's rays on Earth are more direct in June?

Highlight (Yellow) | Location 1329

"How does white light coming from the Sun turn blue as it passes through the atmosphere?"

Highlight (Yellow) | Location 1343

attacks by religion on science are vastly more common than the reverse,

Highlight (Yellow) | Location 1344

Recently in Georgia, the school board wanted to put a disclaimer sticker on biology textbooks. But you do not find scientists asking for disclaimers to be affixed to Bibles in church.

Highlight (Yellow) | Location 1347

in the famous Scopes "monkey" trial,* the science teacher lost the case.

Highlight (Yellow) | Location 1373

Mars was once an oasis of running water. It's bone dry today. And there is no evidence that it will ever return to its previous state. Same for Venus, on which there's a runaway greenhouse effect, leaving the surface at 900 degrees Fahrenheit.

Highlight (Yellow) | Location 1375

you cannot ignore all that does not cycle, select what does, and declare Yin and Yang to be an operational principle of our universe.

Highlight (Yellow) | Location 1395

I lament the loss of useful philosophers that predated modern physics—Immanuel Kant, David Hume, Kurt Gödel, Bertrand Russell, Ernst Mach. Not coincidentally, the transition to uselessness began when our experiments revealed aspects of the universe that no longer followed what anyone would call common sense. The tenets of relativity and quantum mechanics, for example.

Highlight (Yellow) | Location 1416

think of a professor who faces you from the front of the room; who makes eye contact with the audience; who has invested time and energy thinking about how you think; who pays attention to your attention span; who is aware of what words you know and what words or concepts confuse you; who knows the demographics of the audience—age, gender, nationality, ethnicity, political leanings, cultural leanings, propensity to laugh, to cry; who carries some pop-culture fluency, for easy reference and analogy, but only when teaching the subject can be assisted by such references. That person is not lecturing to you. That person has opened conduits tailored to that audience in that moment, and at that time. That’s communicating.

Highlight (Yellow) | Location 1425

you’ll know when you have perfected the task, that’s when under-informed people come up to you and say, “you’re a natural at this.”

Highlight (Yellow) | Location 1431

Pathos A plaintive appeal to emotions that already reside within us.

Highlight (Yellow) | Location 1447

physics-inspired approach of sifting for the fundamental drivers of a cause or phenomenon—stripped of all ornament. To get there, one must assess how and when to ignore the surrounding details, which can give the illusion of importance, yet in the end, are often irrelevant distractions to solutions of otherwise intractable problems.

Highlight (Yellow) and Note | Location 1525

evolutionary biologist Richard Dawkins. He notes that we who die are the lucky ones. Most people—most genetic combinations of who could ever exist—will never be born, and so will never have the opportunity to die.

Furthering my thought that i started life by wining the lottery when i was conceived

Highlight (Yellow) | Location 1609

In some ways, we define our culture (passively or actively) by what we do as a nation as expressed through congressional funding priorities.

Highlight (Yellow) | Location 1643

the Homo sapiens genome is capable of generating trillions of unique humans, which means most people who could ever exist will never even be born.

Highlight (Yellow) | Location 1644

So death is a kind of privilege of those few of us who have known life. Such a cosmic perspective empowers me to celebrate every day I am alive. And I share it with you, as a kind of scientific solace on the life and death of your loved ones.

Highlight (Yellow) | Location 1747

The towers were so tall that for someone on the observation deck, the horizon was forty-five miles away. This distance was far enough along Earth's curved surface for the Sun to set two minutes later for the person on the observation deck than it did for someone on the ground floor.

Highlight (Yellow) | Location 1782

one must always recognize the difference between knowing that something is true, knowing that something is not true, and not knowing one way or another. It's the not knowing part that leaves singular events susceptible to inventive accounts

Highlight (Yellow) | Location 1785

conspiracy theorists know the answers before they investigate, which contaminates their analysis—seducing them to accept what supports their thesis and reject or ignore or not notice that which conflicts with it.

Highlight (Yellow) | Location 1840

In the first century AD, the famous astronomer and mathematician Claudius Ptolemy felt that way as he studied the motions of the planets against the background stars and penned: "When I trace at my pleasure the windings to and fro of the Heavenly Bodies, I no longer touch earth with my feet. I stand in the presence of Zeus himself, and take my fill of ambrosia."[†]

Highlight (Yellow) | Location 1857

The list . . . 1.The Bible " . . . to learn that it's easier to be told by others what to think and believe than it is to think for yourself" 2.The System of the World by Isaac Newton " . . . to learn that the universe is a knowable place." 3.On the Origin of Species by Charles Darwin " . . . to learn of our kinship with all other life on Earth." 4.Gulliver's Travels by Jonathan Swift " . . . to learn, among other satirical lessons, that most of the time humans are Yahoos." 5.The Age of Reason by Thomas Paine " . . . to learn how the power of rational thought is the primary source of freedom in the world." 6.The Wealth of Nations by Adam Smith " . . . to learn that capitalism is an economy of greed, a force of nature unto itself." 7.The Art of War by Sun Tzu " . . . to learn that the act of

killing fellow humans can be raised to an art.” 8.The Prince by Machiavelli “. . . to learn that people not in power will do all they can to acquire it, and people in power will do all they can to keep it.”

Highlight (Yellow) | Location 1877

When you are under the influence of dogma, you say, do, and think what others tell you to. And that’s always easier than thinking for yourself or resisting the powers that established the dogma in the first place.

Highlight (Yellow) | Location 1900

I teach my children about all the world’s major religions. Not pejoratively but anthropologically, which is, I think, a sensible way to broach comparative religion. In this way, they know that while there are multiple belief systems in the world regarding God and gods, there is only one science, and that science is the same no matter your birthplace, on Earth or anywhere else in the cosmos.

Highlight (Yellow) | Location 1909

a story about the horseshoe that hung in Niels Bohr’s office. This famous physicist was asked why he, a man of science, believed in such superstitious things. He’s rumored to have replied, “They tell me it works even if you don’t believe in it.”

Highlight (Yellow) | Location 1934

Science, on the other hand, is a system of knowing and discovery that is decoupled from your nationality or birthplace or ancestry or politics or who you worship. It’s a system of knowing the natural world that is immune to opinion, but not to experiment.

Highlight (Yellow) | Location 1948

I try to spend twice as much time to make things half as long. You don’t want to be guilty of a Shakespearean, “Methinks the Gentlemen doth protest too much.”

Highlight (Yellow) | Location 1950

adage, “When an argument last longer than five minutes then both sides are wrong.”

Highlight (Yellow) | Location 1965

you never want to win an argument for being a better writer, or because you have a bigger vocabulary than your opponent. In this way, the strength of the argument rests on the strength of the argument rather than the strength of your literacy.

Highlight (Yellow) | Location 1990

there is an almost-infinite number of ways you can combine numbers to get other numbers.

Highlight (Yellow) | Location 1993

this information did not exist before the attacks, thereby missing an opportunity to predict them. That's because after-the-fact numerological deductions can be made for any event of any date of any year (you would just combine the numbers in another, equally sensible way), yet the results appear to the eye and the mind as a magical, even mystical connection.

Highlight (Yellow) | Location 1998

numerology is granting access to entertainment rather than access to reality.

Highlight (Yellow) | Location 2023

Galileo's dictum§: "In my mind God wrote two books. The first book is the Bible, where humans can find the answers to their questions on values and morals. The second book of God is the book of nature, which allows humans to use observation and experiment to answer our own questions about the universe."

Highlight (Yellow) | Location 2027

Galileo was a religious man, but was nonetheless compelled to say: "I do not feel obliged to believe that the same God who has endowed us with senses, reason and intellect has intended us to forego their use and by some other means to give us knowledge which we can attain by them. He would not require us to deny sense and reason in physical matters which are set before our eyes and minds by direct experience or necessary demonstrations."

Highlight (Yellow) | Location 2068

no engineer, in his or her right mind, would ever design a system that ingested liquids, solids, communicated, and respiration through the same orifice.

Highlight (Yellow) | Location 2070

Good design is obvious when it's there—the opposable thumb. Stereo-vision. Speech. Ball sockets (shoulders and hips). The shape and strength of our skull, to name a few. But you are in denial of bad design, not because it is not there, but because it falls outside of your religious philosophy, and are thus blind to it.

Highlight (Yellow) | Location 2082

Natural selection never claims perfect design, or even good design, only a design that is more effective than that of a competing species, allowing survival long enough to reproduce. Nothing else matters to the process.

Highlight (Yellow) | Location 2088

perhaps the deepest religious question of them all—If there is no God, then how can life have meaning? Who should care if you ever lived? Why would it matter if any one of us were Stalin or Einstein? Hitler or Mother Teresa?

Highlight (Yellow) | Location 2094

looked within yourself? By doing so, it's not hard to find meaning in life by doing meaningful things—caring for others less fortunate than you, raising children, accomplishing hard tasks that give you physical, intellectual, or emotional fulfillment. The drive to do this, without any reference to religious texts at all, can be quite fulfilling.

Highlight (Yellow) | Location 2103

My motivation is simply to create meaning and self worth—not only in my own life, but in the lives of others.

Highlight (Yellow) | Location 2121

Kairos A propitious moment for decision or action.

Highlight (Yellow) | Location 2199

as often as I can, I seek out people smarter than I am to talk to and hang out with. My wife, for example, has a PhD in mathematical physics. She knows a ton more stuff than I do about all kinds of things. And I won't have it any other way.

Highlight (Yellow) | Location 2296

So if ambition can be found in almost any package, and since by design, there are more people who do not get A's than those who do, then perhaps somebody (a dedicated educator) ought to be in search of those who have it. Or better yet, design curricula that develop it.

Highlight (Yellow) | Location 2410

I do not worry about what they know as much as I worry about how they think.

Highlight (Yellow) | Location 2412

Teaching someone how to think is hard, and takes more effort on the part of the teacher and student. Among other things, it encourages them to ask questions. It involves being comfortable with ignorance, if that happens to be our collective state of knowledge at the time. It involves experiment and inquiry.

Highlight (Yellow) | Location 2418

When their flashlight doesn't work, I do not say, "It needs more batteries," I say, "Let's test the batteries to see if they are dead." We then put the batteries in a battery tester to investigate first-hand. When they ask me a question that I do not know, I reply, "Let's find out," and we go to a book or surf the Internet for answers. If they believe something in the absence of evidence, I ask them, "Why do you believe this?" or "How do you know this?"

Highlight (Yellow) | Location 2422

right now, my daughter is transitioning out of the tooth fairy stage. She now thinks the tooth fairy has been her parents all these years. This prompted a major discussion in her class. So they proposed an experiment to test the idea. The next person whose tooth came out would not notify his or her parents, but would simply take home the tooth and put it under the pillow. Any real tooth fairy would know about this. But parents would not. If there's no money in the morning, then the experiment strongly argues against the existence of a tooth fairy. That is an example of how to think mattering more than what to think.

Highlight (Yellow) | Location 2429

in the era of experiments, when a theory has been resoundingly supported by data—that theory does not one day, all of a sudden, become wrong. The worst that can happen to the theory is that it gets incorporated into a larger, more powerful idea of how the universe works. So the Big Bang is here to stay, in its current form or in a form where it plays a part of a larger cosmic understanding.

Highlight (Yellow) | Location 2499

atheist Jews will commonly celebrate the high holidays with no less ritual than practicing Jews, right on down to leaving an open seat at the Seder table for Elijah, and making sure the front door is unlocked, so he can just walk right in if he happens to show up. Why would an atheist Jew do this? The answer is not hard. Rituals and traditions account for some of the strongest binding forces among peoples of the world.

Highlight (Yellow) | Location 2506

best bet might be to not enforce the literalism of anything religious, but to keep him plugged into the beautiful traditions of the religion, and emphasize the value of ritual as a seed and taproot of community.

Highlight (Yellow) | Location 2508

embrace the value of love and compassion for people and for relationships.

Highlight (Yellow) | Location 2515

2003 60mm Meade brand refracting telescope.

Highlight (Yellow) | Location 2519

copy of H.A. Rey's The Stars†—the first astronomy book my dad gave me, way back in 1955.

Highlight (Yellow) | Location 2525

There's nothing like the right telescope in the right hands of the right person at the right time for the right price.

Highlight (Yellow) | Location 2547

continually felt your guidance ahead of me, your support behind me, and your love beside me.

Highlight (Yellow) | Location 2601

B.o.B. and the Flat Earth The popular hip-hop artist B.o.B. (“bee oh bee”) is vocal about his flat-Earth beliefs. Early in 2016 he took to social media with his ideas. Normally I would ignore such a foray, but he got my attention when he claimed that the laws of math and physics show that Earth is flat. In the geekiverse, those are fighting words. Friday, January 27, 2016 Video Letter to the Rapper B.o.B. Delivered on The Nightly Show with Larry Wilmore, Comedy Central

Highlight (Yellow) | Location 2638

Climate change deniers are misinformed. But so are people who think vaccines gives you autism. And so are people who think genetically modified foods are bad for you. These science-denying postures cross political boundaries.

Highlight (Yellow) | Location 2678

So, I don’t quite know what to say to people who react explosively in the face of objective truths, attacking the person who delivers the information. But what’s clear is that we now live in a world where differences of opinion lead to fights rather than conversations.

Highlight (Yellow) | Location 2686

I remain convinced of Shakespeare’s aphorism: “A rose by any other name would smell as sweet.”

Highlight (Yellow) and Note | Location 2697

I’m talking about ignorant bloopers, like reversing the sunset to pretend you have filmed a sunrise. These are not time-symmetric events. Are cinematographers too sleepy to wake up before the Sun and get the real footage?

Should both rise and set up/down and to right, if N of Tropic of Cancer, like FL.

Highlight (Yellow) | Location 2707

What about those majestic stars in the ceiling of New York City’s Grand Central Terminal? Rather than just admitting that the backwards constellations were a mistake, a sign in the lobby during renovation tells us, “Said to be backwards, [the ceiling is] actually seen from a point of view outside our solar system.” But a second error has now been committed in an attempt to cover up the first: no point of view in our galaxy will reverse the constellation patterns of Earth’s night sky. As you leave the solar system, and travel among the stars, all that happens to Earth’s constellations is that they become scrambled and wholly unrecognizable.

Highlight (Yellow) and Note | Location 2736

I've never forgotten the story of your high school gym teacher, who highlighted your body-type as one that would not make a good runner in the track and field unit of class. Your reaction? "Nobody is going to tell me what I cannot do with my life." You immediately took up running. You also ran in Hitler's Berlin Stadium for the 1946 "GI Olympics." The post-war world was not yet ready for a traditional Olympics, so this special event contested soldier-athletes of the various theaters of conflict around the world. And by college you became world class in middle distance races, at one time capturing the fifth fastest time in the world for the 600-yard run.

My Brown prof...

Highlight (Yellow) | Location 2759

Striving to do what is right, without regard to who takes notice, should be a model for us all.

Highlight (Yellow) | Location 2760

stories and perspectives about navigating the people, politics, funding streams, and legacies of institutions deeply informed my (successful) efforts to create, from whole cloth, a brand new Department of Astrophysics at the American Museum of Natural History. You taught me that in life, it's not good enough to be right. You also must be effective.
